CSE563 — Unit II Numerical Worksheet

Vectors and Matrices

40 questions • CO2 • Solutions Booklet

Instructor Note

Full worked solutions for all 40 questions, with the standard arithmetic checks shown (trace/determinant for eigenvalues, $AA^{-1} = I$ for inverses, weights summing to 1 for bilinear interpolation). Numbering matches the student worksheet exactly.

A. Vectors (Q1–Q7)

Q1. For $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (1, 4, -2)$, compute $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - 3\mathbf{v}$ and $\mathbf{u} \cdot \mathbf{v}$.

Solution. $\mathbf{u} + \mathbf{v} = (4, 3, 0)$. $2\mathbf{u} = (6, -2, 4)$, $3\mathbf{v} = (3, 12, -6)$, so $2\mathbf{u} - 3\mathbf{v} = (\mathbf{3},\ \mathbf{-14},\ \mathbf{10})$. $\mathbf{u} \cdot \mathbf{v} = 3 - 4 - 4 = \mathbf{-5}$.

Q2. Find $\|\mathbf{a}\|$ and the unit vector along $\mathbf{a} = (2, -3, 6)$.

Solution. $\|\mathbf{a}\| = \sqrt{4 + 9 + 36} = \sqrt{49} = \mathbf{7}$. Unit vector $= \left(\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right) \approx (0.286,\ -0.429,\ 0.857)$. *Check:* $\frac{4+9+36}{49} = 1$ ✓.

Q3. Find the angle between $\mathbf{u} = (1, 1, 0)$ and $\mathbf{v} = (1, 0, 1)$.

Solution. $\mathbf{u} \cdot \mathbf{v} = 1$. $\|\mathbf{u}\| = \|\mathbf{v}\| = \sqrt{2}$.

$$\cos\theta = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = \mathbf{60}^\circ.$$

Q4. Determine whether $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 5, 1)$ are orthogonal. If not, find the projection of $\mathbf{a}$ onto $\mathbf{b}$.

Solution. $\mathbf{a} \cdot \mathbf{b} = 2 - 5 + 3 = \mathbf{0}$, so they **are orthogonal**. Consequently the projection of $\mathbf{a}$ onto $\mathbf{b}$ is the **zero vector** — $\mathbf{a}$ casts no shadow along $\mathbf{b}$.

Q5. Find the projection of $\mathbf{u} = (4, 3)$ onto $\mathbf{v} = (1, 0)$, and interpret the result geometrically.

Solution.

$$\text{proj}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v} = \frac{4}{1}(1, 0) = (\mathbf{4}, \mathbf{0}).$$

Since $\mathbf{v}$ is the unit vector along the x-axis, the projection is simply the x-component of $\mathbf{u}$ — the shadow of $\mathbf{u}$ on the horizontal axis.

Q6. Determine whether $(1, 2, 1)$, $(2, 1, 0)$, $(1, -1, -1)$ are linearly independent.

Solution.

$$\det\begin{pmatrix}1 & 2 & 1\\ 2 & 1 & 0\\ 1 & -1 & -1\end{pmatrix} = 1(1\cdot(-1) - 0\cdot(-1)) - 2(2(-1) - 0(1)) + 1(2(-1) - 1(1))$$

$$= 1(-1) - 2(-2) + 1(-3) = -1 + 4 - 3 = \mathbf{0}.$$

Determinant is zero, so the vectors are **linearly dependent**. Indeed $(1,-1,-1) = (2,1,0) - (1,2,1)$.

Q7. Compute $\mathbf{u}\times\mathbf{v}$ for $\mathbf{u} = (1,0,2)$, $\mathbf{v} = (0,3,1)$, and verify the result is orthogonal to both.

Solution.

$$\mathbf{u}\times\mathbf{v} = \big(0(1) - 2(3),\ 2(0) - 1(1),\ 1(3) - 0(0)\big) = (\mathbf{-6},\ \mathbf{-1},\ \mathbf{3}).$$

Check: $(-6,-1,3)\cdot(1,0,2) = -6 + 0 + 6 = 0$ ✓; $(-6,-1,3)\cdot(0,3,1) = 0 - 3 + 3 = 0$ ✓.

B. Matrices (Q8–Q17)

Q8. For $A = \begin{pmatrix}3 & 1\\ 2 & 4\end{pmatrix}$ and $B = \begin{pmatrix}1 & -1\\ 0 & 2\end{pmatrix}$, compute AB and BA.

Solution.

$$AB = \begin{pmatrix}3+0 & -3+2\\ 2+0 & -2+8\end{pmatrix} = \begin{pmatrix}3 & -1\\ 2 & 6\end{pmatrix}, \qquad BA = \begin{pmatrix}3-2 & 1-4\\ 0+4 & 0+8\end{pmatrix} = \begin{pmatrix}1 & -3\\ 4 & 8\end{pmatrix}.$$

$AB \neq BA$ — multiplication is not commutative.

Q9. Compute A^2 for $A = \begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}$, and state $\det(A^2)$ without further multiplication.

Solution.

$$A^2 = \begin{pmatrix}4+0 & 2+3\\ 0+0 & 0+9\end{pmatrix} = \begin{pmatrix}4 & 5\\ 0 & 9\end{pmatrix}.$$

$\det A = 6$, and $\det(A^2) = (\det A)^2 = \mathbf{36}$. *Check directly:* $4(9) - 5(0) = 36$ ✓.

Q10. Evaluate $\det\begin{pmatrix}1 & 2 & -1\\ 3 & 0 & 2\\ -1 & 4 & 1\end{pmatrix}$.

Solution. Expand along row 1:

$$\det = 1(0\cdot 1 - 2\cdot 4) - 2(3\cdot 1 - 2\cdot(-1)) + (-1)(3\cdot 4 - 0\cdot(-1))$$
$$= 1(-8) - 2(5) - 1(12) = -8 - 10 - 12 = \mathbf{-30}.$$

Q11. Find A^{-1} for $A = \begin{pmatrix}4 & 7\\ 2 & 6\end{pmatrix}$ and verify $AA^{-1} = I$.

Solution. $\det A = 24 - 14 = 10$.

$$A^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}.$$

Check: $AA^{-1} = \begin{pmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ ✓.

Q12. Determine whether $A = \begin{pmatrix} 3 & 6 \\ 2 & 4 \end{pmatrix}$ is invertible. Justify.

Solution. $\det A = 12 - 12 = \mathbf{0}$, so A is **singular (not invertible)**. Reason: row 2 is $\frac{2}{3}$ of row 1, so the rows are linearly dependent and $\operatorname{rank} A = 1 < 2$.

Q13. Solve $\begin{cases} 3x + 2y = 16 \\ x - y = 2 \end{cases}$ using the matrix inverse.

Solution. $A = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}$, $\det A = -3 - 2 = -5$, so $A^{-1} = \frac{1}{-5}\begin{pmatrix} -1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 0.2 & 0.4 \\ 0.2 & -0.6 \end{pmatrix}$.

$$\mathbf{x} = A^{-1}\begin{pmatrix} 16 \\ 2 \end{pmatrix} = \begin{pmatrix} 3.2 + 0.8 \\ 3.2 - 1.2 \end{pmatrix} = \begin{pmatrix} \mathbf{4} \\ \mathbf{2} \end{pmatrix}.$$

Check: $3(4) + 2(2) = 16$ ✓; $4 - 2 = 2$ ✓.

Q14. Solve $\begin{cases} x + 2y = 8 \\ 3x - y = 3 \end{cases}$ using Cramer's rule.

Solution. $\det A = \det \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = -1 - 6 = -7$.

$$x = \frac{\det \begin{pmatrix} 8 & 2 \\ 3 & -1 \end{pmatrix}}{-7} = \frac{-8 - 6}{-7} = \frac{-14}{-7} = \mathbf{2}, \qquad y = \frac{\det \begin{pmatrix} 1 & 8 \\ 3 & 3 \end{pmatrix}}{-7} = \frac{3 - 24}{-7} = \frac{-21}{-7} = \mathbf{3}.$$

Check: $2 + 6 = 8$ ✓; $6 - 3 = 3$ ✓.

Q15. Find the eigenvalues of $A = \begin{pmatrix} 6 & -2 \\ -2 & 9 \end{pmatrix}$ and verify them using the trace and determinant.

Solution. $\operatorname{tr} A = 15$, $\det A = 54 - 4 = 50$.

$$\lambda^2 - 15\lambda + 50 = 0 \Rightarrow (\lambda - 10)(\lambda - 5) = 0 \Rightarrow \lambda_1 = \mathbf{10},\ \lambda_2 = \mathbf{5}.$$

Check: $10 + 5 = 15 = \operatorname{tr}$ ✓; $10 \times 5 = 50 = \det$ ✓.

Q16. Find the eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, and confirm the eigenvectors are orthogonal.

Solution. tr $= 6$, det $= 8$: $\lambda^2 - 6\lambda + 8 = 0 \Rightarrow \lambda = \mathbf{4}, \mathbf{2}$.
$\lambda = 4$**:** $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = v_2$, so $\mathbf{v}_1 = (1,1)$. $\lambda = 2$**:** $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = -v_2$, so $\mathbf{v}_2 = (1,-1)$.
Orthogonality: $(1,1)\cdot(1,-1) = 0$ ✓— guaranteed because A is symmetric. Normalised: $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$.

Q17. Find the eigenvalues of the triangular matrix $A = \begin{pmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{pmatrix}$.

Solution. For a triangular matrix $\det(A - \lambda I)$ is the product of the diagonal entries of $A - \lambda I$:

$$(2-\lambda)(3-\lambda)(7-\lambda) = 0.$$

Eigenvalues: $\lambda = 2, 3, 7$ — simply the diagonal entries. *Check:* sum $= 12 = \operatorname{tr} A$ ✓; product $= 42 = \det A$ ✓.

Q18. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, verify that $(A^T)^{-1} = (A^{-1})^T$.

Solution. $\det A = -2$, so $A^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$, giving $(A^{-1})^T = \begin{pmatrix} -2 & 1.5 \\ 1 & -0.5 \end{pmatrix}$.
$A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$, $\det A^T = -2$, so $(A^T)^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1.5 \\ 1 & -0.5 \end{pmatrix}$.
The two agree ✓.

C. Principal Component Analysis (Q18–Q23)

Q19. The data are $(2,1), (4,3), (6,5)$. Compute the mean vector and the centred data matrix.

Solution. $\bar{x} = 12/3 = 4$, $\bar{y} = 9/3 = 3$. Mean vector $= (\mathbf{4}, \mathbf{3})$. Centred: $(-2,-2), (0,0), (2,2)$. *Check:* each centred column sums to zero ✓.

Q20. For the centred data of the previous question, compute the 2×2 covariance matrix and its eigenvalues. Comment on what the result says about the data.

Solution. $\sum \tilde{x}^2 = 4 + 0 + 4 = 8$; $\sum \tilde{y}^2 = 8$; $\sum \tilde{x}\tilde{y} = 4 + 0 + 4 = 8$. With $n - 1 = 2$:

$$S = \frac{1}{2}\begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}.$$

$\operatorname{tr} S = 8$, $\det S = 16 - 16 = 0$, so $\lambda^2 - 8\lambda = 0$, giving $\lambda_1 = \mathbf{8}$, $\lambda_2 = \mathbf{0}$.
Interpretation: the second eigenvalue is exactly zero, so the data have *no* variance perpendicular to the first component — all three points lie exactly on a straight line. PC1 explains 100% of the variance and one dimension suffices.

Q21. Compute the covariance matrix of $(1,2),(2,4),(3,6),(4,8)$ and find both eigenvalues.

Solution. $\bar{x} = 2.5$, $\bar{y} = 5$. Centred: $(-1.5,-3),(-0.5,-1),(0.5,1),(1.5,3)$. $\sum \tilde{x}^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$; $\sum \tilde{y}^2 = 9+1+1+9 = 20$; $\sum \tilde{x}\tilde{y} = 4.5 + 0.5 + 0.5 + 4.5 = 10$. With $n-1 = 3$:

$$S = \frac{1}{3}\begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix} = \begin{pmatrix} 1.6667 & 3.3333 \\ 3.3333 & 6.6667 \end{pmatrix}.$$

$\operatorname{tr} S = 8.3333$, $\det S = 11.111 - 11.111 = 0$.

$$\lambda_1 = \mathbf{8.3333}, \qquad \lambda_2 = \mathbf{0}.$$

Again perfectly collinear ($y = 2x$), so PC1 carries 100% of the variance.

Q22. Given $S = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, find both eigenvalues, the first principal component (normalised), and the percentage of variance it explains.

Solution. $\operatorname{tr} S = 7$, $\det S = 10 - 4 = 6$: $\lambda^2 - 7\lambda + 6 = 0 \Rightarrow \lambda_1 = \mathbf{6}$, $\lambda_2 = \mathbf{1}$.
PC1 ($\lambda = 6$): $(5-6)w_1 + 2w_2 = 0 \Rightarrow w_1 = 2w_2$. Take $(2,1)$; $\|(2,1)\| = \sqrt{5}$.

$$\mathbf{w}_1 = \left(\tfrac{2}{\sqrt{5}},\ \tfrac{1}{\sqrt{5}}\right) \approx (0.8944,\ 0.4472).$$

Variance explained: $6/7 = \mathbf{85.7}\%$.

Q23. Using $\mathbf{w}_1 = (0.8944, 0.4472)$ from the previous question, project the centred point $(3,-1)$ onto PC1 and onto PC2 $= (-0.4472, 0.8944)$. Verify that the squared projections sum to the squared length of the point.

Solution.

$$z_1 = 3(0.8944) + (-1)(0.4472) = 2.6832 - 0.4472 = \mathbf{2.2360}.$$

$$z_2 = 3(-0.4472) + (-1)(0.8944) = -1.3416 - 0.8944 = \mathbf{-2.2360}.$$

Check (Parseval / rotation preserves length): $z_1^2 + z_2^2 = 5.0 + 5.0 = 10$, and $\|(3,-1)\|^2 = 9 + 1 = 10$ ✓. Projecting onto a complete orthonormal basis loses nothing; discarding PC2 is what makes PCA lossy.

Q24. A PCA on a 5-feature dataset yields eigenvalues $12.0, 5.5, 1.8, 0.5, 0.2$. How many components are needed to retain at least 90% of the variance?

Solution. Total $= 12.0 + 5.5 + 1.8 + 0.5 + 0.2 = 20.0$.

PC	λ	Individual	Cumulative
1	12.0	60.0%	60.0%
2	5.5	27.5%	87.5%
3	1.8	9.0%	**96.5%**
4	0.5	2.5%	99.0%
5	0.2	1.0%	100%

Two components give only 87.5%, which falls short. **Three components** are needed (96.5%), reducing the dimensionality from 5 to 3.

D. Linear Discriminant Analysis (Q24–Q27)

Q25. $C_1 = \{(2,3),(3,4),(4,5)\}$ and $C_2 = \{(6,2),(7,3),(8,4)\}$. Compute both class means and the between-class mean difference $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$.

Solution. $\boldsymbol{\mu}_1 = \left(\frac{2+3+4}{3}, \frac{3+4+5}{3}\right) = (\mathbf{3}, \mathbf{4})$. $\boldsymbol{\mu}_2 = \left(\frac{6+7+8}{3}, \frac{2+3+4}{3}\right) = (\mathbf{7}, \mathbf{3})$. $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (\mathbf{-4},\ \mathbf{1})$.

Q26. For the data of the previous question, compute S_1, S_2 and S_W.

Solution. C_1 deviations from $(3,4)$: $(-1,-1),(0,0),(1,1)$. $\sum d_x^2 = 2$, $\sum d_y^2 = 2$, $\sum d_x d_y = 1+0+1 = 2$, so $S_1 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$.

C_2 deviations from $(7,3)$: $(-1,-1),(0,0),(1,1)$ — identical, so $S_2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$.

$$S_W = S_1 + S_2 = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}.$$

Note: $\det S_W = 16 - 16 = 0$, so S_W is *singular* — each class lies exactly on a line of slope 1, giving zero scatter perpendicular to it. Standard LDA cannot be applied directly here; in practice one regularises by adding εI.

Q27. C_1 has $\boldsymbol{\mu}_1 = (1,2)$, C_2 has $\boldsymbol{\mu}_2 = (5,6)$, and $S_W = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$. Find the LDA direction $\mathbf{w}$ and explain why it is not simply parallel to $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$.

Solution. $S_W^{-1} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.125 \end{pmatrix}$ and $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = (-4,-4)$.

$$\mathbf{w} = S_W^{-1} \begin{pmatrix} -4 \\ -4 \end{pmatrix} = \begin{pmatrix} -2 \\ -0.5 \end{pmatrix} \propto (\mathbf{4},\ \mathbf{1}).$$

The mean difference points along $(1,1)$, but the classes are four times more spread in y than in x (scatter 8 vs 2). LDA therefore **down-weights the noisy y direction** and tilts the axis toward x, where the classes are tighter and separation is more reliable. Dividing by the within-class scatter is exactly what distinguishes LDA from simply joining the means.

Q28. Using $\mathbf{w} = (4,1)$, $\boldsymbol{\mu}_1 = (1,2)$ and $\boldsymbol{\mu}_2 = (5,6)$ from the previous question, compute the decision threshold and classify the new point $(3,4)$.

Solution. Projected means: $\mathbf{w}^T\boldsymbol{\mu}_1 = 4(1) + 1(2) = 6$; $\mathbf{w}^T\boldsymbol{\mu}_2 = 4(5) + 1(6) = 26$.

$$\text{Threshold} = \frac{6 + 26}{2} = \mathbf{16}.$$

New point: $\mathbf{w}^T(3,4) = 12 + 4 = 16$.
The projection equals the threshold exactly, so $(3,4)$ lies **on the decision boundary** and LDA is indifferent between the two classes. (Any tie-breaking rule — e.g. assign to the class with the larger prior — must be stated explicitly.)

E. Image Transformation (Q28–Q35)

Q29. Write the 3×3 homogeneous matrix that translates by $(5, -2)$, and apply it to $(3, 7)$.

Solution.

$$T = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \qquad T \begin{pmatrix} 3 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 5 \\ 1 \end{pmatrix}.$$

Image: $(8, 5)$**.**

Q30. Write the matrix that scales by $(3, 0.5)$ about the origin and apply it to the square with vertices $(0, 0), (2, 0), (2, 2), (0, 2)$.

Solution.

$$S = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Images: $(0, 0) \to (0, 0)$; $(2, 0) \to (6, 0)$; $(2, 2) \to (6, 1)$; $(0, 2) \to (0, 1)$. The 2×2 square becomes a 6×1 rectangle. Area goes from 4 to 6, a factor of $\det S = 1.5$ ✓.

Q31. Rotate the point $(4, 0)$ by $60°$ anticlockwise about the origin. Give the exact and decimal answer.

Solution. $\cos 60° = 0.5$, $\sin 60° = \frac{\sqrt{3}}{2} \approx 0.8660$.

$$x' = 4(0.5) - 0(0.866) = \mathbf{2}, \qquad y' = 4(0.866) + 0(0.5) = 2\sqrt{3} \approx \mathbf{3.464}.$$

Check: $\|(2, 3.464)\| = \sqrt{4 + 12} = 4$ — rotation preserves length ✓.

Q32. A point is first rotated by $90°$ anticlockwise about the origin, then translated by $(2, 3)$. Find the single composite matrix and apply it to $(1, 0)$.

Solution. Rotate first, translate second, so $M = T\,R$ (right-most acts first):

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Apply to $(1, 0, 1)^T$: $x' = 0 - 0 + 2 = 2$; $y' = 1 + 0 + 3 = 4$. **Image:** $(2, 4)$**.** *Step check:* $(1, 0) \xrightarrow{R} (0, 1) \xrightarrow{T} (2, 4)$ ✓.

Q33. Repeat the previous question with the operations in the opposite order (translate by $(2, 3)$ first, then rotate by $90°$). Compare the two results.

Solution. Now $M' = R\,T$:

$$M' = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -3 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Apply to $(1,0)$: $x' = 0 - 0 - 3 = -3$; $y' = 1 + 0 + 2 = 3$. **Image:** $(-3, 3)$.
Comparison: $(2,4)$ versus $(-3,3)$ — completely different. The rotation acts on the translation offset in the second case, spinning it to a new position. *Order of transformations matters*: matrix multiplication is not commutative.

Q34. Find the single matrix that rotates by $90°$ anticlockwise about the point $(2,1)$, and apply it to $(4,1)$.

Solution. $M = T(2,1)\,R(90°)\,T(-2,-1)$:

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 3 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Apply to $(4,1)$: $x' = -1+3 = 2$; $y' = 4-1 = 3$. **Image:** $(2,3)$. *Geometric check:* $(4,1)$ is 2 units east of the centre $(2,1)$; a $90°$ anticlockwise turn sends east to north, giving $(2, 1+2) = (2,3)$ ✓.

Q35. Write the matrix for a shear of the x-coordinate by factor $k = 2$, and find the image of the unit square $(0,0), (1,0), (1,1), (0,1)$. What is the area of the image?

Solution.

$$\mathrm{Sh} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (x, y) \mapsto (x + 2y,\ y).$$

Images: $(0,0) \to (0,0)$; $(1,0) \to (1,0)$; $(1,1) \to (3,1)$; $(0,1) \to (2,1)$. The square becomes a parallelogram. $\det(\mathrm{Sh}) = 1$, so the **area is unchanged at 1** — shear slants a shape without altering its area.

Q36. A transformation is given by $M = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$. Find M^{-1} and use it to recover the original point whose image is $(9,5)$.

Solution. M scales by 2 then translates by $(3,-1)$. Its inverse translates by $(-3,1)$ then scales by $\frac{1}{2}$:

$$M^{-1} = \begin{pmatrix} 0.5 & 0 & -1.5 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{pmatrix}.$$

Apply to $(9,5,1)^T$: $x = 4.5 - 1.5 = \mathbf{3}$; $y = 2.5 + 0.5 = \mathbf{3}$. *Forward check:* $(3,3) \to$ scale $\to (6,6) \to$ translate $\to (9,5)$ ✓.

F. Image Interpolation (Q36–Q40)

Q37. Pixel values are $I(0,0) = 50$, $I(1,0) = 100$, $I(0,1) = 150$, $I(1,1) = 200$. Estimate $I(0.5, 0.5)$ by bilinear interpolation.

Solution. $a = b = 0.5$, so all four weights equal 0.25 (sum = 1 ✓).

$$I = 0.25(50 + 100 + 150 + 200) = 0.25(500) = \mathbf{125}.$$

At the exact centre bilinear interpolation returns the plain average of the four corners.

Q38. For the same four pixels, estimate $I(0.8, 0.3)$ by bilinear interpolation. Show the four weights and verify they sum to 1.

Solution. $a = 0.8$, $b = 0.3$.

Corner	Weight	Contribution
$I(0,0) = 50$	$(1-a)(1-b) = 0.2 \times 0.7 = 0.14$	7.0
$I(1,0) = 100$	$a(1-b) = 0.8 \times 0.7 = 0.56$	56.0
$I(0,1) = 150$	$(1-a)b = 0.2 \times 0.3 = 0.06$	9.0
$I(1,1) = 200$	$ab = 0.8 \times 0.3 = 0.24$	48.0
	$\sum = 1.00$ ✓	**120.0**

Two-step check: along x at y=0: $50 + 0.8(50) = 90$; at y=1: $150 + 0.8(50) = 190$; then along y: $90 + 0.3(100) = 120$ ✓.

Q39. For the pixels of Q36, give the nearest-neighbour estimate at $(0.8, 0.3)$ and state the difference from the bilinear answer.

Solution. Round each coordinate: $(0.8, 0.3) \to (1, 0)$. Nearest-neighbour value = **100**. Bilinear gave 120, so the difference is **20** grey levels. Nearest neighbour ignores the three other corners entirely, which is why it produces blocky output but preserves exact original pixel values.

Q40. The 1-D row $[10, 50, 30, 70]$ is enlarged to 7 samples. Give the nearest-neighbour and linear-interpolated outputs.

Solution. Output positions map back to source coordinates $0, 0.5, 1, 1.5, 2, 2.5, 3$.

Output	Source	Nearest	Linear
0	0.0	10	10
1	0.5	50	$10 + 0.5(40) = 30$
2	1.0	50	50
3	1.5	30	$50 + 0.5(-20) = 40$
4	2.0	30	30
5	2.5	70	$30 + 0.5(40) = 50$
6	3.0	70	70

Nearest: $[10, 50, 50, 30, 30, 70, 70]$. **Linear:** $[10, 30, 50, 40, 30, 50, 70]$. (Half-way ties rounded up for the nearest-neighbour column.)

Q41. A 4×4 image is to be rotated by $30°$. Explain with a calculation why interpolation is unavoidable, using the destination pixel $(3, 1)$ and inverse mapping about the origin.

Solution. Inverse mapping applies $R(-30°)$ to the destination coordinate to find where it came from:

$$x = 3\cos(-30°) - 1\sin(-30°) = 3(0.8660) + 1(0.5) = 3.098,$$

$$y = 3\sin(-30°) + 1\cos(-30°) = -1.5 + 0.866 = -0.634.$$

The source location is $(3.098,\ -0.634)$ — **not an integer pixel position**, and in this case it even falls outside the image (negative y), which is why rotated images need both interpolation *and* a background fill policy.

Since no pixel exists at a fractional coordinate, the intensity must be *estimated* from the surrounding integer pixels — by nearest neighbour, bilinear or bicubic. This is unavoidable for any rotation that is not a multiple of 90°.